

Worksheet · Section 5.5–5.6

Legal & Ethical Concerns · Safe Computing



The Big Picture

Know who owns digital content and how to share it legally, and how to protect yourself online. Practice licenses, plagiarism, common attacks, and encryption.

Part 1 · Vocab Match

Write the letter of the matching definition on each line.

- | | |
|--------------------------------------|--|
| _____ 1. Intellectual property | A. An unauthorized wireless network that can intercept data |
| _____ 2. Creative Commons | B. Information that can identify a person, like a Social Security number |
| _____ 3. Open source | C. Presenting someone else's work as your own |
| _____ 4. Plagiarism | D. Creations of the mind, like code, music, or art, owned by the creator |
| _____ 5. PII | E. Licenses that let creators share their work under conditions |
| _____ 6. Phishing | F. Software designed to damage a computer or take control of it |
| _____ 7. Malware | G. Code that anyone can freely use and modify |
| _____ 8. Keylogging | H. Requires two or more types of evidence to log in |
| _____ 9. Encryption | I. Encoding data so only authorized people can read it |
| _____ 10. Public key encryption | J. Secretly recording every keystroke |
| _____ 11. Multifactor authentication | K. Uses a public key to encrypt and a private key to decrypt |
| _____ 12. Rogue access point | L. Tricking a user into giving personal information |

Part 2 · Fill It In

Use each word from the word bank once.

PII

public

two

updates

credit

open source

private

copyright

1. A song you write is protected by _____.
2. Creative Commons “BY” licenses require you to give the creator _____.
3. Code that anyone can freely use and modify is _____.
4. Your Social Security number is _____.
5. To send Bob a secret message, encrypt it with Bob's _____ key.
6. Only Bob's _____ key can decrypt that message.
7. Multifactor authentication requires at least _____ types of evidence.
8. Installing software _____ fixes security weaknesses.

Name:

Class:

Date:

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Practice

Part 3 · Name the Attack

Check the best name for each threat.

Item	Phishing	Malware	Keylogging	Rogue access point
1. An email says “verify your account” with a fake link				
2. A free game secretly installs software that takes control of your computer				
3. A program secretly records everything you type				
4. A fake “Airport_Free_WiFi” network set up by an attacker				
5. A text says you won a prize and asks for your card number				
6. A virus copies itself into other files				

Part 4 · Legal & Ethical?

Scenario	OK or not OK? Why?
Using a CC BY image with credit in a school project	_____
Copying an essay from a website and turning it in	_____
Modifying open-source code and citing it in your Create Task	_____
Selling an app that uses a CC BY-NC song	_____
Downloading a paid movie from a pirate site	_____

Part 5 · Encryption Lab

Caesar cipher, shift 3 (A→D, B→E, ..., X→A).

Encrypt “SAFE” → _____ Decrypt “KHOOR” → _____

Is a Caesar cipher symmetric or public key?

_____ Why is it weak?

Public key: Alice sends Bob a secret. She encrypts with _____;

he decrypts with _____.

Part 6 · Privacy Audit

List three pieces of PII an app might collect, explain how they could be combined to learn more about you, and name two settings or habits that protect you.

**Think:**

Public key = an open padlock you hand to everyone. Anyone can snap it shut on a message, but only YOUR private key opens it.

Name: _____

Class: _____

Date: _____

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Exam Practice

Part 7 · Exam Practice

★ Warm-up

1. Which is personally identifiable information?

- A. A random number
- B. The weather forecast
- C. A public library's hours
- D. Your home address

★★ On Level

2. Which is an example of multifactor authentication?

- A. A long password
- B. A username only
- C. A password plus a fingerprint scan
- D. Two different passwords

★★ On Level

3. A photo has a CC BY-NC license. What does this allow?

- A. Any use without credit
- B. Claiming the photo as your own
- C. Selling the photo
- D. Use with credit, for non-commercial purposes only

★★★ Exam Hard

4. What is the role of a certificate authority?

- A. It blocks all websites
- B. It issues certificates that verify who owns an encryption key
- C. It writes viruses
- D. It stores everyone's passwords

★★★ Exam Hard

5. Which TWO actions protect against phishing? (*Select two.*)

- A. Replying with your password to confirm your identity
- B. Checking the sender and the link's real address before clicking
- C. Turning off multifactor authentication
- D. Going to the company's website directly instead of using the link

Part 8 · Written Response

Explain how public key encryption lets you shop safely online without ever sharing a secret key with the store in advance.
